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Advancing circularity in timber construction: Design for disassembly and reuse, and innovative wood-based connections

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ABSTRACT: This paper investigates circularity in timber construction, focusing on extending wood's lifespan through Design for Disassembly & Reuse and innovative timber connections. It examines the feasibility of circular practices to address resource conservation and climate mitigation. The study analyzes case studies, including connections with birch plywood gusset plates with threaded wood dowels, reclaimed timber portal frames, and a circular timber bridge concept. The paper aims to provide architects and engineers with a practical approach based on simple principles for promoting circularity in timber construction. It also offers reflections on design strategies, balancing functionality, aesthetics, and structural requirements.

1 INTRODUCTION

1.1 *Motivation*

The global climate crisis and the increased demand and competition for renewable resources, such as wood, require changes in how timber is used in construction. It is projected that the demand for wood will continue to increase in the coming years and might surpass the supply available with sustainable forest management (Mantau et al. 2010). As a renewable material, wood has large potential to facilitate more sustainable building practices. However, traditional timber production and construction methods do not fully utilize this potential. They often lead to limited lifecycles of wood-based materials, produce too much waste and use resources inefficiently (Pramreiter et al. 2023). To address these issues, the timber construction industry has started shifting from a conventional linear economy towards a more circular-economy approach (Cristescu et al. 2020). What circularity entails and how it can be applied in practice is, however, not straightforward.

1.2 *Circular timber construction*

Circularity in timber construction refers to strategically managing wood resources to maintain their solid state for as long as possible. This involves designing buildings with an emphasis on material recovery and reuse, enabling timber components to be disassembled and reassembled with minimal damage. In essence, the goal is to close the loop by keeping wood in circulation rather than allowing it to degrade or be discarded prematurely.

An efficient approach to facilitate circularity is through Design for Disassembly and Reuse (DfDR). Disassembly (or sometimes Deconstruction) is the process of dismantling a building or structure into parts that can be utilized again. Reuse refers to the utilization of the recovered construction elements without recycling them into new products. DfDR is meant to guide architects and engineers to create buildings and structures where components can be easily dismantled, reused, or repurposed (Mofatt & Russell 2001, Guy & Ciarimboli 2006). In

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DfDR for timber construction, innovations in connection methods play a critical role. By utilizing reversible, non-destructive and repetitive connections, the longevity of wooden building components is significantly increased. Moreover, by reducing the need for metallic elements in connections, the CO_2 emissions can be significantly reduced.

1.3 *Purpose, scope, and limitations*

This paper aims to explore the potential and feasibility of circular practices in timber construction, focusing on DfDR and novel wood-based connections. By presenting relevant case studies, we exemplify and analyze how circularity principles can be effective while meeting structural requirements and architectural aspects.

Three specific case studies are described: 1) timber joints with birch plywood gusset plates, 2) reclaimed timber portal frames, and 3) a circular wood-based bridge concept. Each case study represents innovative work developed by the authors to illustrate the feasibility of circular strategies. These examples offer practical insights into the design and engineering processes behind circular timber construction.

The analysis of the case studies is based on previously stated principles. Additionally, the outcomes of the study are formed into practical recommendations to promote circularity. While the scope of circularity in timber construction is broad, this paper focuses on specific cases. By narrowing the discussion, we aim to offer detailed insights into DfDR in timber construction.

2 METHODS

The paper combines theory and practice by outlining four circularity principles which are used to evaluate three built outcomes. The principles are developed through state-of-the-art literature and intuitive synthesis by the authors, all of whom are engaged professionally with the topic. The built outcomes encompass laboratory testing, a full-scale demonstrator, and a completed commercial project that fulfills the requirements of a program.

These are all focused on the use of timber in contemporary construction, including timber joints utilizing birch plywood gusset plates, portal frames from reclaimed timber using wooden dowel joints, and a timber bridge with modular parts and reversible connections. As design projects, each of the case studies is materialized through broad criteria, even if their purpose is precisely defined and executed.

3 CIRCULARITY IN TIMBER CONSTRUCTION

3.1 *Previous studies*

In construction, circularity involves creating systems where material waste can be prevented (or significantly reduced) and materials can be reused, recycled, or recovered rather than discarded. In the context of timber construction, circularity emphasizes the reuse of wood-based materials, reducing waste during construction, and designing structures that can be easily disassembled and reassembled, rather than demolished. A key framework for conceptualizing extended lifetimes for wood material is biomass cascading (Vis et al. 2016). This theorizes how wood material can occupy varied durations across multiple products and applications. By design, this highlights the importance of acknowledging the requirements of future cascades during the development of early applications for wood, favoring long-term products, maintaining material integrity, and enabling reversibility for future applications.

Assessments of DfDR potential during design cover a broad territory, from cultural practices to technical requirements, and predictions of future concerns and priorities (Munch-Petersen 2020). This makes DfDR challenging to assess explicitly, relying on design principles that are precedented in offering extended material lifetimes.

Crowther (2005) examines historical examples to identify recurring patterns in DfDR. By translating these patterns into common aspects the author provides a comprehensive list of underlaying DfDR principles: 1) Minimize different component types, 2) Use mechanical connections, 3) Use open building system, 4) Design modularly, 5) Avoid specialized equipment, 6) Separate structure from cladding, 7) Ensure access to all parts, 8) Size components

for easy handling, 9) Provide handling and positioning methods, 10) Allow realistic tolerances, 11) Minimize number of connectors, 12) Minimize connector types, 13) Design joints/components for repeated use, 14) Enable parallel disassembly, 15) Label component types, 16) Use prefabrication and mass production, 17) Use lightweight materials/components, 18) Mark disassembly points, 19) Retain all information on components and materials. These principles are deemed relevant for various levels of circularity, including material recycling, component remanufacturing, component reuse, and even building relocation.

A more comprehensive list of principles is provided by Guy & Ciarimboli (2008). According to them the 10 key principles of Design for Disassembly are: 1) Document materials and deconstruction methods, 2) Select materials cautiously, 3) Design accessible connections, 4) Minimize chemical connections, 5) Use bolts, screws, and nails, 6) Separate mechanical, electrical and plumbing systems, 7) Design with labor ease in mind, 8) Keep structure and form simple, 9) Ensure interchangeability, 10) Prioritize safe deconstruction. It should be noted that they focus more on disassembly than reuse. The above principles are arranged in groups by Cristescu et al. (2020) as follows: Timely disassembly, Separability of parts, Minimizing safety risks (these three contribute to efficient Disassembly), Simple form and structure, and Interchangeability (these two facilitate Reuse).

In the current study we focus on principles that extend the lifetime of timber structures. The discussion and evaluation of the three case studies later presented in this paper are based on four main principles, namely *Residue minimization*, *Repetitiveness in design*, *Reversibility of connections*, and *Repairability of construction elements*. These we call the "4R principles" and explain in the following subsections.

3.2 *Residue minimization. How well is wood's integrity maintained for and during DfDR?*

Wood is a subtractive material during its processing from tree to product, highlighting the need to consider the forms and formats of material in evaluations of cascading efficacy (Pramreiter et al. 2023). Wood's value generally decreases as it passes through cascades; therefore, it is important to reduce contaminants, both physical and chemical, which are key for enabling wood to be valued across more than one application (Niu et al. 2021).

3.3 *Repetitiveness. How consistent and uniform are part types within a structure?*

Repetition of part geometries and functions is a common principle for DfDR, since it reduces the complexity of handling component inventories through modularity (Brütting et al. 2019). This should not be confused with the standardization of final structural solutions. Repetition proves promising not only for reassembling structures in the same form, but also for giving rise to new structures from existing parts since part functionality may be agnostic across consecutive applications (Retsin 2019). Furthermore, discrete systems are theorized to enable continuous reconfiguration of structural components to cater to changing spatial requirements and forms, as well as for parts to be exchanged in the circumstance of failure or maintenance (Kunic & Naboni 2023).

3.4 *Reversibility. How does the design of joints facilitate their reversibility?*

This paper departs from the study of joints by Much-Petersen (2020), who organizes different joint types on a linear spectrum according to how intuitively their reversibility can be understood (Figure 1). Joints at the intuitive end of the spectrum are considered to hold greater reversibility potential, since they mitigate the need for supporting documentation. This highlights how DfDR of joints requires careful consideration for the information available to a future scenario, and that joint types vary in how they are perceived, regardless of their actual reversibility in perfect practice.

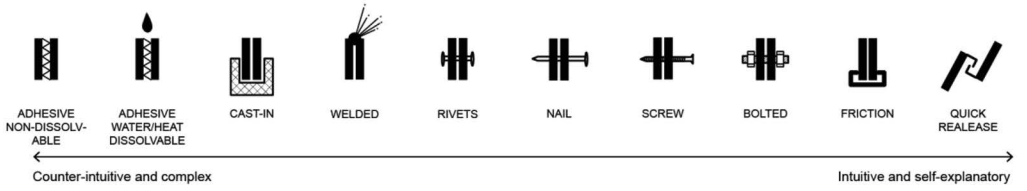


Figure 1. Reversibility of joints after Munch Petersen (2020).

3.5 *Repairability. How easy is it to detach and replace damaged parts?*

Modern constructions are designed to last for a given number of years (design lifetime), typically 50 to 100 years. Repairability or replacements of load-bearing parts is in general not part of the design strategy in common modern structures. Although it is commonly not the main concern for many types of structures, the possibility of repairability or replacements of structural parts might become very relevant for outdoor constructions.

Here, the load-bearing structure could be designed as several separate elements or members, rather than a single massive piece. The advantage of using compound members is that repair or replacement of a damaged member would be performed without exaggerated efforts, which results in an enhanced longevity of the construction as a whole. The choice of compound members (instead of massive members) requires, however, the adoption of more complicated connections that both allow for a quick and easy replacement of damaged parts and provide the structure with adequate lateral stiffness.

4 CASE STUDIES

4.1 *General*

The following subsections present three case studies that demonstrate the use of the 4R principles to assess the circularity of design projects. The case studies represent the authors' work, where the 4R approach provides simple qualitative circularity self-assessment. This easy method, although not comprehensive, might help structure discussions during collaborative projects and compare various alternative concepts. This contribution illustrates the concept whereas a more detailed study will be presented in a subsequent research paper.

4.2 *Timber joint design with reduced utilization of fossil-based materials*

Conventional metallic plates in modern timber connections have negative implications regarding environmental impact. Wood-based joint plates and, when possible, wood-based fasteners are ideal parts of timber structures, not only for their eco-friendliness but also due to other advantages, such as better workability.

Plywood is one of the most common types of wood-based structural panels. Compared to steel plates, plywood plates are competitive in structural connections due to their low carbon footprint, high tolerance during assembly, good fire resistance, relatively low cost, and less prefabrication demand. Among the cons of using plywood in timber connections instead of steel plates, one can mention: the larger thickness required, and less predictable failure modes that might prevent the engineer from choosing this material.

However, plywood made of birch veneers possesses superior strength and stiffness performance compared to most softwood plywood types making it promising for structural use. The proper understanding of the mechanical properties under various loading conditions is crucial for a safe and proper design of connections. Several investigations regarding birch plywood's mechanical properties and the possibility of being used in timber connections have been performed (Sæby 2024, Wang et al 2024). Results show that this solution can be successfully used in connections, especially for timber structures of small-to-medium sizes. Figure 2 shows an example of a specimen in a testing rig. The concept has its own implications on DfDR, detailed here in the context of the four Rs:

- Residue minimization: Plywood is relatively high-level utilization of wood in the material cascade. Additionally, using plywood in the primary load-bearing system could be seen as an upcycling of the material, keeping it in this state for a longer time.
- *Repetitiveness*: the connections if used in larger structures, e.g. in large-span trusses or frame corners, could be standardized to facilitate reuse after building deconstruction.
- *Reversibility*: if undamaged, these connections are fully reversible. It is however important to keep the loads below the level that leads to irreversible deformation of the components.
- *Repairability*: Spliced connections with mechanical fasteners and simple geometry are generally easier to dismantle and replace than complex geometries that include glueing and/or welding.



Figure 2. Test setup of a frame corner with birch plywood and wood pegs (Sæby, 2024).

4.3 *Reversible timber-only connections and Wood ReFramed*

The building segment *Wood ReFramed* was realized to demonstrate novel load bearing systems for reclaimed timber (Browne & Larsen 2024). The project seeks to facilitate extended lifetimes for post-consumer wood in high-value structural applications. This is achieved by adopting a variant of the so-called HB-system (developed by the Swedish engineer Hilding Brosenius), which maintains material integrity by mitigating the need for planing and precise part interfacing as well as being an efficient system for longer spans (Pajunen et al. 2024).

The structure consists of a series of portal frames, each with the same external geometry, with inner web members allocated according to their structural requirements. The components utilize multi-shear connections using wooden dowels, presenting a low-impact solution for a structural typology with a large quantity of connections. The 4Rs are articulated in the project as follows:

- *Residue minimization*: The system is made up of small, somewhat irregular pieces which ensure a high level of material integrity and keeps the value of wooden parts that would have been used for non-structural purposes otherwise.
- *Repetitiveness*: The series of frames strikes a balance between uniformity and tailored relationships between parts and performance. The 45-degree inner web elements are all uniform in length. However, their cross-sections vary according to the inherited heterogeneity of a stock of secondary timber. This enables a variety of material forms to be thought of as discrete where their variations are maintained within rectilinear components.
- *Reversibility*: Recent studies have demonstrated that dowel connections can be reversed by calibrating their geometric changes with different humidities (Browne & Larsen 2023). Exposing assembled components to drier environments causes dowels to shrink, allowing them to be removed for reorganizing parts according to new functional requirements, or for complete disassembly for other applications. This conveys the potential of material properties for facilitating reversibility, as well as geometric relations and part uniformity.
- *Repairability*: even if not purposefully addressed in the project, the small, disintegrated pieces might be easy to replace if needed. This might be relevant for the bottom of the column parts that could be exposed to impact from various sources.



Figure 3. Details of the Wood ReFramed segments (photos: Sara Lohse).

4.4 *Demountable timber bridge BIFROST*

The BIFROST timber-based bridge concept has been developed with sustainability in mind. Started as innovation competition in October 2022, a real bridge was completed in March 2024 in the locality of Uddebo in Sweden. A picture of the bridge is shown in Figure 4 (left).

The main design principles were to a) enhance the use of wood and minimize structural weight, b) ensure easy assembly and maintenance, and c) emphasize lightness and naturality in the architectural expression. As an effort to minimize environmental impact and promote sustainable resource use, the concept integrates circular design principles, notably DfDR to extend the service life of bridges, reduce waste, and create a more resilient, low-carbon infrastructure solution.

In traditional construction, bridges are often built in a way that makes deconstruction labor-intensive, costly, and damaging to materials. BIFROST addresses this by a design that allows for easy dismantling of the structure and possible reuse of its components in other projects. The key circularity principles mentioned in section 3 are represented in BIFROST as follows:

- *Residue minimization*: This aspect is only partly addressed in the project. The load bearing timber parts are made of glulam, which is processed wood. Furthermore, the vertical elements are impregnated to protect them from weather. The elements of the arch and the horizontal stabilizing members, however, are made of untreated glulam beams that are covered with wooden panels on the sides and thin metal sheets on the top. These compromises are deemed necessary to ensure sufficient durability of the bridge. The bridge deck is made of trapezoidal metal sheeting screwed to wooden planks. Although the metal sheets represent abiotic materials, they could potentially be reused after the bridge's service life.
- *Repetitiveness*: The bridge employs a modular design with standardized units and dimensions. It ensures that parts can be easily removed and replaced when necessary, prolonging the bridge's service life. It also allows for flexibility in design, making it easier to adapt the structure to new environments or reassemble it in different configurations. Also, the connections are somewhat repetitive i.e., the same connection is used at various locations in the structure. This feature adds additional flexibility and adaptability for future reuse and repurposing of the structure or parts of it. An example of a typical connection is shown in Figure 4 (right) joining two glulam beams. The double door hinge-like connectors able to transfer bending moments, can be used for beams with various heights, and different angles between the parts can be achieved with the same connectors.
- *Reversibility*: This aspect is, to a certain extent, addressed at the connections. They typically make use of dowels rather than welding or glueing. Such joints can be dismantled without

damaging the timber elements, allowing for the recovery and reuse of components. This approach not only simplifies repairs and upgrades but also enables entire sections of the bridge to be reused in new constructions. It should be mentioned, however, that there are several welded details at the steel parts of the joints itself, as well as in the steel tension tie of the arch.

– *Repairability*: Ease of maintenance and repair was a key point from the beginning in the development of the concept. Building on previous experience with damage to timber bridges, each wooden part is designed in a way that possible biological degradation (typically rot) does not spread to larger areas. Damaged timber members are easy to replace due to their limited size and weight (all are below 4 m) and mechanical connections. Furthermore, they are made of locally available products, allowing for timely replacement. The load-bearing wooden parts are protected from weather exposure to further enhance longevity of the structure and reduce repair needs. Unfortunately, the repair and replacement of the steel parts are more cumbersome. To minimize the risk for early degradation they have been galvanized after welding.



Figure 4. View of the bridge in Uddebo, built based on the BIFROST concept (left) and its demountable connection(right).

5 DISCUSSION

The timber examples presented in the text discussed innovative timber joint types utilizing birch plywood instead of steel plates, a demonstrator of novel load-bearing system for reclaimed timber and a full-scale commercial bridge structure. All examples in different scales – from joint to full-scale structure, had an ambition to advance circularity in timber construction. That has been achieved by the application of Design for Disassembly and Reuse (DfDR) strategy.

In reflecting upon the examples, it became evident that all examples largely embraced the 4R principles: Residue minimization, Repetitiveness, Reversibility and Repairability. These enable longer life spans for timber projects fostering circularity.

Although not discussed in the paper, which focused on structural factors, the proposed 4R principles offer further opportunities, but also challenges that should be considered from project to project. These include:

– Functionality/usability of designs,
– Tectonic expression and aesthetics.

Although the DfDR strategy and 4R principles offer a tangible approach for fostering circular timber construction, it is important to apply them in conjunction with design quality offering functionality and appropriate space use. These together with the tectonic and aesthetic expression have a crucial role for a building's social acceptance which influences care and longevity.

Therefore, for advancing circularity in timber construction it is important to carry out critical analysis and evaluation of design solutions, on a project-by-project basis. By evaluating architectural (cultural, aesthetic, tectonic expression) and balancing structural/functional requirements will provide a basis for long-lasting circular timber structures.

The three examples are clearly very different, yet they all, within their own scale, offer an example of possible good practice through DfDR and the application of the 4R principles.

6 CONCLUSIONS

The three examples present a possible strategy that through DfDR may extend timber structures' circularity. To fully benefit from the approaches further research needs to be carried out mapping technical/structural and architectural aspects through further cases and built examples. Understanding both the opportunities as well as the challenges better and applying the knowledge onto further projects will foster circularity in timber construction. This is a clear and very important goal in addressing the climate crisis and building better with wood.

Achieving a more sustainable building practice through long-lasting high-quality solutions and circular approaches is something to aim for. It is hoped that this paper presents a small step towards this goal.

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